TITS-ALGEBRA COMPATIBILITY CRITERION FOR REAL-UNIVERSAL SEMISIMPLE RATIONAL LIE ALGEBRAS

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Editorial note. The task counts the final semisimple direct-sum criterion only. The existence of some universal rational form(Proposition1.2, old item2007), separate classical-type classifications and the restriction-of-scalars proposition are not additional counted claims.

Editorial note. In927, the second factor is printed with lambda_i again. The preceding923–924 and following931–933 explicitly use lambda_i and lambda_j. Both real factors are H under the hypotheses, so this duplicated index does not change the calculation; preserve the original formula and disclose the index.

Editorial note. The source explicitly writes in912 that only the forward direction is displayed and that the argument is reversible. The allowed-input boundary does not supply the converse as a theorem. The solver still has to check all global dominant weights: star-fixed weights give products of the common quaternion class, and non-star-fixed weights have center L. For the latter case, the Tits classes of an already R-universal outer factor split over its inner field L, using the retained root-lattice invariance and weight-center reasoning461–492. This is the scope of the author compressed reversibility claim; it is flagged for independent completeness review rather than adding an alternate authored proof.

Editorial note. When comparing Tits algebras after tensor products, tilde denotes Brauer equivalence; D itself is the division representative. A product H tensor H is matrix algebra/Brauer trivial, not a division algebra. The source uses that distinction in921–933.

Editorial note. The quaternionic type-D classification720–765, including the unresolved pair of real half-spin classes in759, is not an input or part of this task. Conditions(1) and the retained general representation criterion concern actual R-universal factors, not that case table.

Editorial note. All of415–493 remains a local proof obligation. Only the external standard Tits results and basic Brauer/Galois facts listed below may be freely invoked.

Editorial scope. Keep the entire author proof, including the explicitly compressed reverse direction. Separate notes expose its required case split and the duplicated index; no replacement proof or formula is inserted.

Allowed standard prerequisites (editorial).

- Tits highest-weight descent and Tits division algebra theorem. External classical classification of irreducible F-representations, the center as the fixed field of the star stabilizer, and the corresponding Brauer class. Do not give RunivIff or SameFixed for free.
- Classical representation and Galois-cohomology foundations. Endomorphism algebras commute with scalar extension; dominant weights can be

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shifted into the root lattice; the star action has finite image. The current descent and common-field argument must be proved.
- Brauer theory over Q, R and C. R division algebras with these centers are R/C/H; H tensor H is Brauer trivial, quaternion classes have order2, a central division algebra is determined by its Brauer class, and Tits classes add with central characters. No semisimple compatibility criterion is included as an input.

Proposition 0.1. *A direct sum $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$ of simple Lie algebras over $\mathbb{Q}$ is $\mathbb{R}$-universal if and only if:*

(1) *each $\mathfrak{g}_i$ is $\mathbb{R}$-universal,*
(2) *either $\mathfrak{g}$ is inner, or it becomes inner over an imaginary quadratic extension L of $\mathbb{Q}$, and the $*$-action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$ is the same as the $*$-action of $\mathrm{Gal}(L/\mathbb{Q})$.*
(3) *we have $D_{\mathfrak{g}_i,\mathbb{Q}}(\lambda_i) \cong D_{\mathfrak{g}_j,\mathbb{Q}}(\lambda_j)$ whenever λ_i and λ_j are $*$-invariant dominant weights of G_i and G_j, such that $D_{\mathfrak{g}_i,\mathbb{Q}}(\lambda_i) \neq \mathbb{Q}$ and $D_{\mathfrak{g}_j,\mathbb{Q}}(\lambda_j) \neq \mathbb{Q}$, and*
(4) *either $\mathfrak{g}$ is inner, or it becomes inner over an imaginary quadratic extension L of $\mathbb{Q}$, such that $D_{\mathfrak{g}_i,\mathbb{Q}}(\lambda_i)$ splits over L, for every dominant weight λ_i of every $\mathfrak{g}_i$ that is inner (over $\mathbb{Q}$).*

Definition 0.2. Let $\mathfrak{g}_{\mathbb{Q}}$ be a Lie algebra over $\mathbb{Q}$. (All Lie algebras and all representations are assumed to be finite-dimensional.)

(1) $\mathfrak{g}_{\mathbb{Q}}$ is *universal for real representations* (or $\mathbb{R}$-*universal*, for short) if every real representation of $\mathfrak{g}_{\mathbb{Q}}$ has a $\mathbb{Q}$-form [6, Definition 7.1]. This means that if $\rho\colon \mathfrak{g}_{\mathbb{Q}} \to \mathfrak{gl}(n,\mathbb{R})$ is any ($\mathbb{Q}$-linear) Lie algebra homomorphism, then there exists $M \in \mathrm{GL}(n,\mathbb{R})$, such that $M\rho(x)M^{-1} \in \mathfrak{gl}(n,\mathbb{Q})$, for every $x \in \mathfrak{g}_{\mathbb{Q}}$.
(2) $\mathfrak{g}_{\mathbb{Q}}$ is a $\mathbb{Q}$-*form* of a real Lie algebra $\mathfrak{g}_{\mathbb{R}}$ if $\mathfrak{g}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathfrak{g}_{\mathbb{R}}$.

1. Proof of the results stated in the introduction

We begin by recalling a result of J. Tits that uses Galois cohomology to characterize the irreducible representations of semisimple algebraic groups over fields that are not algebraically closed. However, we will state this work in the setting of Lie algebras, rather than algebraic groups. (We deal only with semisimple Lie algebras and semisimple groups, and the fields under consideration in this paper are all of characteristic zero, so no difficulties arise in making this translation [1, Proposition 7.3.1(iii), p. 393].)

Definition 1.1 ([11, § 4.2])**.** Suppose $\mathfrak{g}$ is a semisimple Lie algebra over a subfield F of $\mathbb{C}$, and let $\mathbf{G}$ be the corresponding simply connected, semisimple algebraic group over F. It is well known that there is a (unique) quasi-split, simply connected algebraic group $\mathbf{G}^q$ over F, and a 1-cocycle $\xi\colon \mathrm{Gal}(\overline{F}/F) \to \overline{\mathbf{G}}^q$, where $\overline{\mathbf{G}}^q$ is the adjoint group of $\mathbf{G}^q$, such that $\mathbf{G}$ is F-isomorphic to the Galois twist ${}^{\xi}\mathbf{G}^q$. This cocycle determines a cohomology class $[\xi] \in H^1(F;\overline{\mathbf{G}}^q)$.

Letting $Z(\mathbf{G}^q)$ be the center of $\mathbf{G}^q$, the short exact sequence $e \to Z(\mathbf{G}^q) \to \mathbf{G}^q \to \overline{\mathbf{G}}^q \to e$ yields a corresponding long exact sequence of Galois cohomology sets, including a connecting map $\delta_*\colon H^1(F;\overline{\mathbf{G}}^q) \to H^2\big(F;Z(\mathbf{G}^q)\big)$. Hence, we have a cohomology class $\delta_*[\xi] \in H^2\big(F;Z(\mathbf{G}^q)\big)$. (If we took a bit more care to ensure that it is well-defined, this would be the *Tits class* of $\mathbf{G}$ [4, p. 426].)

Now, fix a maximal F-torus $\mathbf{T}$ of $\mathbf{G}^q$ that contains a maximal F-split torus, and suppose λ is a weight of $\mathbf{T}$ that is invariant under the $*$-action of the Galois group $\mathrm{Gal}(\overline{F}/F)$. Then the restriction of λ to $Z(\mathbf{G}^q)$ is a $\mathrm{Gal}(\overline{F}/F)$-equivariant

homomorphism from $Z(\mathbf{G}^q)$ to the group $\boldsymbol{\mu}$ of roots of unity in $\mathbb{C}$, so it induces a homomorphism $\lambda_*\colon H^2\big(F; Z(\mathbf{G}^q)\big) \to H^2\big(F; \boldsymbol{\mu}\big)$. Therefore, we may define

$$\beta_{\mathfrak{g},F}(\lambda) = \lambda_*\delta_*[\xi] \in H^2(F; \boldsymbol{\mu}).$$

Remark 1.2. For a semisimple group $\mathbf{G}$ that is defined over a field F, the definition of the $*$-action of $\mathrm{Gal}(\overline{F}/F)$ on the weights of a maximal torus $\mathbf{T}$ presupposes that $\mathbf{T}$ is defined over F and contains a maximal F-split torus of $\mathbf{G}$ [7, p. 66]. When $\mathbf{G}$ is defined over $\mathbb{Q}$, we will use the $*$-actions of both $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$, so we will assume that $\mathbf{T}$ is defined over $\mathbb{Q}$ (so it is also defined over $\mathbb{R}$) and contains both a maximal $\mathbb{Q}$-split torus and a maximal $\mathbb{R}$-split torus. To see that such a choice of $\mathbf{T}$ is always possible, let $\mathbf{T}_1$ be any maximal $\mathbb{Q}$-split torus of $\mathbf{G}$. Then there is a maximal $\mathbb{Q}$-torus $\mathbf{T}$ of the centralizer $\mathbf{C}_{\mathbf{G}}(\mathbf{T}_1)$ that contains a maximal $\mathbb{R}$-split torus [7, Corollary 2 of Proposition 7.8, p. 418], and it is clear that $\mathbf{T}$ has the desired properties.

Proposition 1.3 ([11, Theorem 7.2 and Lemma 7.4])**.** *Suppose $\mathfrak{g}$ is a semisimple Lie algebra over a subfield F of $\mathbb{C}$, and λ is a dominant weight. Then:*

1. *There is an irreducible representation ${}^F\rho_\lambda\colon \mathfrak{g} \to \mathfrak{gl}(n, F)$, for some n, such that ${}^F\rho_\lambda \otimes_F \mathbb{C}$ has an irreducible summand with highest weight λ. Furthermore, ${}^F\rho_\lambda$ is unique up to isomorphism.*
2. *${}^F\rho_{\lambda_1} \cong {}^F\rho_{\lambda_2}$ if and only if λ_1 and λ_2 are in the same orbit of the $*$-action of $\mathrm{Gal}(\overline{F}/F)$.*
3. *${}^F\rho_\lambda \otimes_F \mathbb{C}$ is irreducible if and only if:*
   (a) *λ is invariant under the $*$-action of $\mathrm{Gal}(\overline{F}/F)$, and*
   (b) *$\beta_{\mathfrak{g},F}(\lambda)$ is the trivial element of $H^2(F; \boldsymbol{\mu})$.*

Furthermore, every F-irreducible representation of $\mathfrak{g}$ is isomorphic to ${}^F\rho_\lambda$, for some dominant weight λ.

Notation 1.4. We assume $\mathfrak{g}$ is a semisimple Lie algebra over a field F of characteristic 0, that λ is a dominant weight, and the other notation of Definition 1.1. Furthermore, we let Z^* be the finite, abelian group of all homomorphisms from $Z(\mathbf{G}^q)$ to $\boldsymbol{\mu}$.

Proposition 1.5 ([11, Corollary 3.5, § 4.2, and Lemma 7.4])**.** *Let L be the center of the algebra $D_{\mathfrak{g},F}(\lambda) := \mathrm{End}_{\mathfrak{g}}({}^F\rho_\lambda)$ (which, by Schur's lemma, is a division algebra). Then:*

(1) *$\mathrm{Gal}(\overline{F}/L) = \{\sigma \in \mathrm{Gal}(\overline{F}/F) \mid \sigma(\lambda) = \lambda\}$ (for the $*$-action of $\mathrm{Gal}(\overline{F}/F)$), and*
(2) *$\beta_{\mathfrak{g},L}(\lambda) = [D_{\mathfrak{g},F}(\lambda)]$, after identifying $H^2(L; \boldsymbol{\mu})$ with the Brauer group of L.*

Definition 1.6. We will call $D_{\mathfrak{g},F}(\lambda)$ the *Tits algebra* of $\mathfrak{g}$ corresponding to the weight λ over F. (However, this name is used in the literature for a slightly different algebra that is Brauer equivalent to $D_{\mathfrak{g},F}(\lambda)$ [4, § 27A, p. 377].)

Lemma 1.7. *If $\mathfrak{g}$ is a semisimple Lie algebra over $\mathbb{Q}$, then the following are equivalent:*

1. *$\mathfrak{g}$ is $\mathbb{R}$-universal.*
2. *${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R} \cong {}^{\mathbb{R}}\rho_\lambda$, for every dominant weight λ.*
3. *$D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R}$ is a division algebra, for every dominant weight λ.*
4. *$D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R} \cong D_{\mathfrak{g},\mathbb{R}}(\lambda)$, for every dominant weight λ.*

Proof. (1 $\Rightarrow$ 2) Let ρ be a $\mathbb{Q}$-form of ${}^{\mathbb{R}}\rho_\lambda$. Then ρ must be $\mathbb{Q}$-irreducible (since $\rho \otimes_{\mathbb{Q}} \mathbb{R} \cong {}^{\mathbb{R}}\rho_\lambda$ is $\mathbb{R}$-irreducible) and

$$\rho \otimes_{\mathbb{Q}} \mathbb{C} \cong (\rho \otimes_{\mathbb{Q}} \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} \cong {}^{\mathbb{R}}\rho_\lambda \otimes_{\mathbb{R}} \mathbb{C}$$

has an irreducible summand with highest weight λ. Therefore $\rho \cong {}^{\mathbb{Q}}\rho_\lambda$.

$(2 \Rightarrow 1)$ ${}^{\mathbb{Q}}\rho_\lambda$ is a $\mathbb{Q}$-form of ${}^{\mathbb{R}}\rho_\lambda$.

$(2 \Leftrightarrow 3 \Leftrightarrow 4)$ We have $\operatorname{End}_{\mathfrak{g}}({}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R}) = D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R}$. So ${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R}$ is irreducible (and hence equal to ${}^{\mathbb{R}}\rho_\lambda$) if and only if $D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R}$ is a division algebra (and hence equal to $D_{\mathfrak{g},\mathbb{R}}(\lambda)$). □

Remark 1.8. It is immediate from Definition 1.1 that if the $*$-invariant weights λ_1 and λ_2 have the same restriction to $Z(\mathbf{G}^q)$ (that is, if they represent the same element of Z^*), then $\beta_{\mathfrak{g},F}(\lambda_1) = \beta_{\mathfrak{g},F}(\lambda_2)$. Therefore, if λ is a dominant weight in the root lattice (so λ is trivial on $Z(\mathbf{G}^q)$), and λ is invariant under the $*$-action of $\operatorname{Gal}(\overline{F}/F)$, then $D_{\mathfrak{g},F}(\lambda) = F$.

Lemma 1.9. *Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{Q}$. If $\mathfrak{g}$ is $\mathbb{R}$-universal, then $\mathfrak{g}$ is inner over some imaginary quadratic extension L of $\mathbb{Q}$, and the $*$-action of $\operatorname{Gal}(L/\mathbb{Q})$ is the same as the $*$-action of $\operatorname{Gal}(\mathbb{C}/\mathbb{R})$.*

Proof. Assume $\mathfrak{g}$ is not inner, for otherwise the desired conclusions are obvious. Let L be the (unique) minimal extension of $\mathbb{Q}$ over which $\mathfrak{g}$ becomes inner, and choose some dominant weight λ that is not fixed by any nontrivial element of $\operatorname{Gal}(L/\mathbb{Q})$. We know that L is the center of $D_{\mathfrak{g},\mathbb{Q}}(\lambda)$ (see Proposition 1.5(1)) and that $D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R}$ is a division algebra (see Lemma 1.7(3)). Since L is not $\mathbb{Q}$, this implies it is an imaginary quadratic extension of $\mathbb{Q}$.

It now suffices to show that every every $\operatorname{Gal}(\mathbb{C}/\mathbb{R})$-invariant dominant weight λ is also $\operatorname{Gal}(L/\mathbb{Q})$-invariant. Suppose not. By replacing λ with an appropriate positive integer multiple, we may assume it is in the root lattice, so $D_{\mathfrak{g},\mathbb{R}}(\lambda) = \mathbb{R}$ (see Remark 1.8). Therefore, we must have $D_{\mathfrak{g},\mathbb{Q}}(\lambda) = \mathbb{Q}$ (see Lemma 1.7(4)). However, this contradicts Proposition 1.5(1), since λ is not $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-invariant. □

Proof. We prove only $(\Rightarrow)$, but the argument is reversible. (We use $\sim$ to denote "is Brauer equivalent to".)

(1) Any representation ρ_i of $\mathfrak{g}_i$ extends to a representation of $\mathfrak{g}$ (that is 0 on the other simple factors), so it is clear that $\mathfrak{g}_i$ must be $\mathbb{R}$-universal.

(2) See Lemma 1.9. (This means that all $\mathfrak{g}_i$ become inner over the same quadratic extension L.)

(3) We have

$$\begin{aligned} D_{\mathfrak{g},\mathbb{R}}(\lambda_i + \lambda_j) &\sim D_{\mathfrak{g},\mathbb{R}}(\lambda_i) \otimes_{\mathbb{R}} D_{\mathfrak{g},\mathbb{R}}(\lambda_j) \\ &\cong \big(D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} \mathbb{R}\big) \otimes_{\mathbb{R}} \big(D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} \mathbb{R}\big) \cong \mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \sim \mathbb{R}. \end{aligned}$$

Since $\mathfrak{g}$ is $\mathbb{R}$-universal, this implies $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i + \lambda_j) \sim \mathbb{Q}$, from which we conclude that $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} D_{\mathfrak{g},\mathbb{Q}}(\lambda_j) \sim \mathbb{Q}$, so $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \cong D_{\mathfrak{g},\mathbb{Q}}(\lambda_j)$ (since these are quaternion algebras).

(4) Let λ_j be a dominant weight of some (outer) simple factor G_j that is not fixed by the $*$-action. Then $\lambda_i + \lambda_j$ is not fixed by the $*$-action, so

$$D_{\mathfrak{g},\mathbb{Q}}(\lambda_i + \lambda_j) = D_{\mathfrak{g},L}(\lambda_i + \lambda_j) \sim D_{\mathfrak{g}_i,L}(\lambda_i) \otimes_L D_{\mathfrak{g}_j,L}(\lambda_j) = D_{\mathfrak{g}_i,L}(\lambda_i) \otimes_L L \cong D_{\mathfrak{g}_i,L}(\lambda_i).$$

Since $D_{\mathfrak{g},\mathbb{R}}(\lambda_i + \lambda_j) = D_{\mathfrak{g},\mathbb{C}}(\lambda_i + \lambda_j) = \mathbb{C}$, and $\mathfrak{g}$ is $\mathbb{R}$-universal, we conclude from Lemma 1.7(4) that $D_{\mathfrak{g}_i,L}(\lambda_i) = L$. Since $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i) \otimes_{\mathbb{Q}} L \sim D_{\mathfrak{g},L}(\lambda_i)$, this means that $D_{\mathfrak{g},\mathbb{Q}}(\lambda_i)$ splits over L. □

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